



COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST

LAMINAR FLOW GAUGE

MODEL DF-240BA

ERROR AND TROUBLESHOOTING MANUAL

1. ERRORS

If an error occurs, the digital indicator will show an error number, such as **E-01**, and the number remains shown there until the power switch is turned off. Check the contents of the error by referring to the following table.

ERR signals are output in the following cases.

Error	Description	Cause	Error Code
Instrument error	EEPROM error	EEPROM malfunction	E-01
	System error	CPU ROM operation faulty	E-02
	RAM error	RAM operation faulty	E-03
Sensor error	Analog input range over *1	Sensor faulty or A/D converter malfunction	E-04
	Sensor output error	Sensor faulty	—
External signal input operation error	Zero adjustment (ZIN) operation error	Zero adjustment was executed when there is a deviation of over ± 50 count (3.5-digit display set) or ± 500 count (4.5-digit display set)	—

*1 **E-04** occurs when the sensor output exceeds its full-scale range when the power is turned on. The measurement mode recovers when the sensor output returns to the full-scale range.

2. TROUBLESHOOTING

Check the following points before ordering to have your device repaired.

Problem	Symptom	Countermeasures
No power is supplied.	No LEDs are lit on.	Check that the power cord was correctly connected.
No key operation is accepted.	When a key is pressed, the alarm beeps.	Check that the keyboard lock was not set. In the setup mode, set Loc or oFF.
	When the ENT key is pressed after a setting has been made in the setup mode, the alarm beeps, and the setting cannot be validated.	Check whether or not [Loc] is set to [on1] or [on2] in the setup mode. If so, set [Loc] to [oFF].
	The CH , AUTO ZERO and/or HOLD keys are invalid.	Have these functions been set using external input? (Settings made in external input have priority over direct key operation.)
No external input (I/O) is accepted.	Any of the external input signals are invalid.	- Check that the connectors were correctly connected. - Check the input voltage of pin 37 (EXTPSIN) of the external I/O connector (DATA I/O). - In 5V input specification, if a voltage higher than 5V is applied to pin 37 (EXTPSIN) of the external I/O connector, the internal circuits may be damaged.
No input from the RS-232C is accepted.	No input from the RS-232C is valid. No response returns even if a command is entered.	- Check that the connectors were correctly connected. - Check that the signal specifications of the host computer are correct. - Check that the baud rate of the host computer agrees with that of the DF-231BA. - Check that the main unit is not in the setup mode. (No measurement is effected in the setup mode, and so no response returns at all.)
	Error (error field: 80) returns.	Was the correct command used? Confirm that the command has been entered completely in the upper case (commands must be entered in all capital letters).
	Error (error field: 10) returns.	Has the setting been made using external input? (External input has priority over the RS-232C.)
	Signal is transmitted although there must be no flow.	- Check for any leak of the fluid from the tube. - Check the sensor connection
Flow rate reading doesn't agree with the actual flow rate.		- Check for leakage from tubing. - Check for contamination in the tubing and the connection port. - Check whether the flow rate is fluctuating widely or exceeding the operation range.